

33 An Introduction to Invertebrates 686

CONCEPT 33.1 Sponges are basal animals that lack tissues 690

CONCEPT 33.2 Cnidarians are an ancient phylum of eumetazoans 691

Medusozoans 692

Anthozoans 693

CONCEPT 33.3 Lophotrochozoans, a clade identified by molecular data, have the widest range of animal body forms 694

Flatworms 694

Rotifers and

Acanthocephalans 697

Lophophorates: Ectoprocts

and Brachiopods 698

Molluscs 699

Annelids 703

CONCEPT 33.4 Ecdysozoans are the most species-rich animal group 705

Nematodes 705

Arthropods 706

CONCEPT 33.5 Echinoderms and chordates are deuterostomes 713

Echinoderms 713

Chordates 715



34 The Origin and Evolution of Vertebrates 718

CONCEPT 34.1 Chordates have a notochord and a dorsal, hollow nerve cord 719

Derived Characters of Chordates 719

Lancelets 720

Tunicates 721

Early Chordate Evolution 722

CONCEPT 34.2 Vertebrates are chordates that have a backbone 722

Derived Characters of Vertebrates 722

Hagfishes and Lampreys 723

Early Vertebrate Evolution 724

CONCEPT 34.3 Gnathostomes are vertebrates that have jaws 725

Derived Characters of Gnathostomes 725

Fossil Gnathostomes 726

Chondrichthyans (Sharks, Rays, and Their Relatives) 726

Ray-Finned Fishes and Lobe-Fins 728



CONCEPT 34.4 Tetrapods are gnathostomes that have limbs 730

Derived Characters of Tetrapods 730

The Origin of Tetrapods 731

Amphibians 731

CONCEPT 34.5 Amniotes are tetrapods that have a terrestrially adapted egg 734

Derived Characters of Amniotes 734

Early Amniotes 735

Reptiles 735

CONCEPT 34.6 Mammals are amniotes that have hair and produce milk 741

Derived Characters of Mammals 741

Early Evolution of Mammals 741

Monotremes 742

Marsupials 743

Eutherians (Placental Mammals) 744

CONCEPT 34.7 Humans are mammals that have a large brain and bipedal locomotion 748

Derived Characters of Humans 748

The Earliest Hominins 748

Australopiths 749

Bipedalism 750

Tool Use 750

Early *Homo* 750

Neanderthals 752

Homo sapiens 753

Unit 6 Plant Form and Function 757

Interview: Dennis Gonsalves 757

35 Vascular Plant Structure, Growth, and Development 758

CONCEPT 35.1 Plants have a hierarchical organization consisting of organs, tissues, and cells 759

Vascular Plant Organs: Roots, Stems, and Leaves 759

Dermal, Vascular, and Ground

Tissues 762

Common Types of Plant Cells 763

CONCEPT 35.2 Different meristems generate new cells for primary and secondary growth 766

CONCEPT 35.3 Primary growth lengthens roots and shoots 768

Primary Growth of Roots 768

Primary Growth of Shoots 769

CONCEPT 35.4 Secondary growth increases the diameter of stems and roots in woody plants 772

The Vascular Cambium and Secondary Vascular Tissue 773

The Cork Cambium and the Production of Periderm 774

Evolution of Secondary Growth 774

CONCEPT 35.5 Growth, morphogenesis, and cell differentiation produce the plant body 775

Model Organisms: Revolutionizing the Study of Plants 776

Growth: Cell Division and Cell Expansion 776

Morphogenesis and Pattern Formation 777

Gene Expression and the Control of Cell Differentiation 778

Shifts in Development: Phase Changes 778

Genetic Control of Flowering 779

